Independent Sets in 3-Uniform 3-Regular Hypergraphs

Jialin He*

Abstract

Balogh, Bollobás, and Narayanan conjectured that a particular quasi-bipartite hypergraph maximizes the number of independent sets in a uniform regular hypergraph. We prove their conjecture for 3-uniform 3-regular hypergraphs. Every such simple hypergraph on n vertices has at most $253^{n/9}$ independent sets, with equality precisely for disjoint unions of the conjectured nine-vertex extremal hypergraph. We also obtain an exponential loss in the number of edges containing no two vertices with identical incident edges. The proof uses an occupancy bound that retains the proportion of these twin vertices, together with an exact certificate for 800 local counting vectors. We prove the partition-function comparison throughout the interval $[2/3, 5/3]$, with an exponential penalty for every vertex outside the conjectured extremal components. Further results treat all positive activities in the linear class and activities at least one in the twin-free and exact-transversal classes.

1 Introduction

An independent set in a hypergraph is a vertex set containing no edge. Write $i(H)$ for the number of independent sets of H, and let

$$Z_H(\lambda) = \sum_{I \text{ independent in } H} \lambda^{|I|}$$

be its partition function at activity $\lambda \geq 0$. In particular, $Z_H(1) = i(H)$. Throughout the paper, hypergraphs are finite and simple. The empty hypergraph satisfies the counting bounds trivially; statements involving occupancy fractions assume that the number of vertices is positive.

For integers $r \geq 2$ and $d \geq 1$, the hypergraph $H_{r,d}$ has d disjoint sets of size $r-1$ and d further vertices. Its edges are all unions of one of these sets and one of the further vertices. Thus $H_{r,d}$ is r-uniform and d-regular, with rd vertices. Balogh, Bollobás, and Narayanan [1, Conjecture 1.1] conjectured that

$$i(H) \leq i(H_{r,d})^{n/(rd)}$$

for every n-vertex r-uniform d-regular hypergraph. They proved the bound for quasi-bipartite hypergraphs. This class admits a vertex set meeting every edge exactly once, with a matching in the link of each vertex in that set.

We prove the conjecture for $r = d = 3$, without a structural assumption on H. The reference hypergraph $H_{3,3}$ consists of three vertices a_1, a_2, a_3 and three disjoint pairs B_1, B_2, B_3, with edges $\{a_i\} \cup B_j$ for $i, j \in \{1, 2, 3\}$. Its partition function is

$$\begin{aligned} P(\lambda) &= (1+\lambda)^6 + \big((1+\lambda)^3 - 1\big)(1+2\lambda)^3 \\ &= 1 + 9\lambda + 36\lambda^2 + 75\lambda^3 + 81\lambda^4 + 42\lambda^5 + 9\lambda^6. \end{aligned}$$

*School of Mathematical Sciences, Key Laboratory of MEA (Ministry of Education), Shanghai Key Laboratory of PMMP, and Nantong Institute for Applied Mathematics and Artificial Intelligence, East China Normal University, Shanghai 200241, China. Emails: jlhe@math.ecnu.edu.cn. Supported in part by the National Natural Science Foundation of China (Grant No. 12601663) and the Science and Technology Commission of Shanghai Municipality (Grant No. 22DZ2229014).

The first term counts sets avoiding all three a_i; once an a_i is selected, each pair B_j contributes at most one vertex. In particular, $i(H_{3,3}) = 253$.

Two distinct vertices are *twins* if they belong to exactly the same edges. For a 3-uniform 3-regular hypergraph H, let $q(H)$ count the edges containing no pair of twins. Put $\kappa = \frac{259}{379500}$.

Theorem 1.1. *Every n-vertex 3-uniform 3-regular hypergraph H satisfies*

$$i(H) \le 253^{n/9} \exp(-\kappa q(H)).$$

Consequently, $i(H) \le 253^{n/9}$, with equality if and only if H is a disjoint union of copies of $H_{3,3}$.

The upper bound does not require $9 \mid n$. The quantity $q(H)$ measures a local obstruction to extremality. Its vanishing does not by itself imply that H is a union of reference components; the equality argument handles the remaining full-book hypergraphs separately.

Sarantis, Tetali, and Zheng [5, Conjectures 1.2(ii) and 1.7] proposed the stronger partition-function comparison

$$Z_H(\lambda) \le Z_{H_{r,d}}(\lambda)^{n/(rd)} \qquad (\lambda \ge 0),$$

as well as a comparison of individual independent-set coefficients. Their results include several degree-two cases and asymptotic bounds under additional structural assumptions. Theorem 1.1 settles the total-count comparison for $(r, d) = (3, 3)$; it does not settle the partition-function or coefficient conjectures for this pair.

For general parameters, the companion manuscript [4] treats activity and coefficient comparisons near both endpoints.

Our weighted comparison holds on an interval independent of the order of the hypergraph. Let $b(H)$ count the vertices not belonging to a connected component isomorphic to $H_{3,3}$.

Theorem 1.2. *There is an absolute constant $c > 0$ such that every n-vertex 3-uniform 3-regular hypergraph satisfies*

$$Z_H(\lambda) \le P(\lambda)^{n/9} \exp(-cb(H)) \qquad \left(\frac{2}{3} \le \lambda \le \frac{5}{3}\right).$$

Consequently, equality in the unpenalized comparison holds precisely for disjoint unions of $H_{3,3}$ throughout this interval.

For any activity in this interval, a logarithmic deficit of $o(n)$ forces all but $o(n)$ vertices to lie in complete reference components. The constant is uniform over the whole interval. Theorem 1.2 still leaves the all-activity conjecture open.

The application of the occupancy method to hypergraphs by Cohen, Perkins, Sarantis, and Tetali [2] provides a starting point for our proof. Conditional distributions on a root and its neighbors give finite collections of local counting polynomials. We allow all residual pair and triple constraints permitted by the degree budget, including those caused by cross-edges. The additional identities used here transfer joint occupation information between a twin pair and each of its incident third vertices. They give a bound that depends on the proportion of twin vertices, which is fixed as the activity varies. Integrating this bound proves Theorem 1.1.

Section 2 defines the local models and their consistency identities. Section 3 proves the counting theorem, including the exact certificate and the equality case. Section 4 gives further weighted bounds. In the linear class the reference bound holds for every positive activity; in the twin-free class it holds for every activity at least one, with a uniform exponential loss. We also prove a partition-function bound and component stability for hypergraphs with an exact transversal, that is, a set meeting every edge in exactly one vertex. Finally, a book expansion transfers a full-book stability estimate to arbitrary cubic 3-graphs and proves Theorem 1.2.

The supplementary programs regenerate the local models and verify their inequalities exactly. The proofs below establish that these models cover every admissible hypergraph. All logarithms are natural.

2 Local models and consistency identities

Fix a 3-uniform 3-regular hypergraph H with $n > 0$ vertices. For $\lambda > 0$, choose an independent set I with probability $\lambda^{|I|}/Z_H(\lambda)$. Its occupancy fraction is

$$\alpha_H(\lambda) = \frac{\mathbb{E}|I|}{n} = \frac{\lambda Z_H'(\lambda)}{nZ_H(\lambda)}.$$

We first record the structural information that survives conditioning on vertices outside a root neighborhood.

Lemma 2.1. *Every nontrivial twin class has exactly two vertices. If there are t twin pairs, their incident edges form t edge-disjoint copies of*

$$\{uva, uvb, uvc\},$$

where a, b, c are distinct and have no twins. In particular, $q(H) = n - 3t \geq 0$.

Proof. Three twins would force every edge through one of them to be the same triple, contrary to simplicity and degree three. A twin pair u, v therefore has three distinct third vertices a, b, c. If, for example, a had a twin, the edge uva would force that twin to be u or v, producing three twins. Distinct twin pairs are disjoint and cannot belong to the same triple. Their incident edges are consequently disjoint. Finally, 3-uniformity and 3-regularity give $e(H) = n$. □

Call the three edges through a twin pair a *book*, its twin vertices its endpoints, and its three other vertices its leaves. A vertex may be a leaf of several books. Write

$$s = \frac{2t}{n}, \qquad 0 \leq s \leq \frac{2}{3}.$$

The link $L(v)$ of a vertex v is the graph whose edges are the pairs ab for which $vab \in E(H)$. It has three distinct edges and, after isolated vertices are omitted, is one of

$$3K_2, \quad P_3 \sqcup K_2, \quad P_4, \quad K_3, \quad K_{1,3}.$$

Here P_j is the path on j vertices. The root v has a twin precisely when its link is $K_{1,3}$; the center of the star is that twin. At a non-twin root, mark every link edge whose endpoints are twins. Such an edge is isolated in the link. Hence only $3K_2$ and $P_3 \sqcup K_2$ can have marked edges, with at most three and one, respectively.

Put $X_v = \mathbf{1}_{\{v \in I\}}$. For distinct vertices v, u, let $d_H(v, u)$ be the number of edges containing both, and define

$$w_v = \sum_{u \neq v} d_H(v, u) X_u, \qquad p_v = \sum_{ab \in E(L(v))} X_a X_b.$$

For every independent set, double counting gives

$$\sum_v w_v = 6|I|, \qquad \sum_v X_v w_v = 2 \sum_v p_v. \tag{1}$$

Indeed, each selected vertex contributes twice through each of its three incident edges to the first sum. For the second identity, each occupied pair in an edge is counted twice on the left and once in the sum of the p_v.

The next identities retain information at both ends of a book–leaf incidence. For a book with endpoints u, u' and a leaf w, let $Y(w, \{u, u'\})$ be one if neither of the two edges through w other than $uu'w$ has both its other vertices in I, and zero otherwise. This is the availability of w when the edge $uu'w$ is ignored. Its definition does not use X_u or $X_{u'}$.

For $y \in \{0,1\}$, define $R_y(v,I)$ as follows. At a twin root v, with twin v', count the leaves w for which $X_v = X_{v'} = 1$, $X_w = 0$, and $Y(w,\{v,v'\}) = y$. At a non-twin root v, take minus twice the number of marked edges uu' of $L(v)$ for which $X_u = X_{u'} = 1$, $X_v = 0$, and $Y(v,\{u,u'\}) = y$. Then

$$\sum_v R_y(v,I) = 0 \qquad (y = 0,1). \tag{2}$$

Each qualifying book–leaf incidence contributes one at each of the two endpoints and minus two at its leaf. These identities hold for each independent set, before taking expectations.

We now describe the conditional models used in the certificate. Fix a root v, let N be its neighbor set, and condition on $I \setminus (N \cup \{v\})$. An edge not containing v imposes a constraint on its intersection with N exactly when all its outside vertices are selected. Constraints of size one force vertices to be absent. Let $U \subseteq N$ be the remaining available vertices, and let $\mathcal{F}$ be the family of constraints of size two or three, with repetitions removed. Edges entirely outside the neighborhood impose no further restriction because the conditioning has positive probability.

For each $i \in N$, the number of distinct constraints containing i is at most $3 - \deg_{L(v)}(i)$. For a uniform enumeration, we retain constraints meeting $N \setminus U$, although they are redundant for subsets of U. The only additional restrictions needed for the enumeration are the following.

- If the root link is a star, its center lies in U and belongs to no constraint.
- Both endpoints of every marked link edge lie in U. Any constraint meeting one of them contains the whole marked pair.

The first restriction follows because all edges through the center contain the root. For the second, the other two edges through an endpoint contain its twin as well, so no singleton constraint can affect it. We enumerate all families of pair and triple constraints satisfying these degree bounds and restrictions, and all admissible sets U. Ignoring the degree consumed by singleton constraints enlarges the list. Every actual conditional model is included, whether or not every enumerated model is realizable by a hypergraph.

Denote such a model by C. Let $\mathcal{S}_C$ consist of the subsets $S \subseteq U$ containing no member of $\mathcal{F}$. For $S \in \mathcal{S}_C$, set

$$w(S) = \sum_{i \in S} \deg_L(i), \qquad p(S) = |\{e \in E(L) : e \subseteq S\}|.$$

Define its five counting polynomials by

$$\begin{aligned} q_0 &= \sum_{S \in \mathcal{S}_C} \lambda^{|S|}, & q_1 &= \sum_{\substack{S \in \mathcal{S}_C \\ p(S)=0}} \lambda^{|S|}, & t_0 &= \sum_{S \in \mathcal{S}_C} p(S)\lambda^{|S|}, \\ w_0 &= \sum_{S \in \mathcal{S}_C} w(S)\lambda^{|S|}, & w_1 &= \sum_{\substack{S \in \mathcal{S}_C \\ p(S)=0}} w(S)\lambda^{|S|}. \end{aligned} \tag{3}$$

The root may be absent for any $S \in \mathcal{S}_C$, and may be present precisely when $p(S) = 0$. Its conditional partition function is therefore $Z_C = q_0 + \lambda q_1$.

The book statistics can also be recovered from C. At a star root, a leaf w has $Y = 1$ precisely when $w \in U$ and no $F \in \mathcal{F}$ containing w has $F \setminus \{w\} \subseteq S$. At a non-star root and a marked link edge e, the value is $Y = 1$ precisely when neither of the other two link edges is contained in S. Thus define r_y by summing R_y over the legal root states $\varepsilon \in \{0,1\}$ and neighbor sets S, with weight $\lambda^{|S|+\varepsilon}$. These are integer polynomials of degree at most seven, and the conditional mean of R_y is r_y/Z_C.

Write $\sigma = 1$ for a star root and $\sigma = 0$ otherwise, and put

$$a = \frac{\lambda q_1}{Z_C}, \qquad d = a - \frac{w_0 + \lambda w_1}{6Z_C}, \qquad h = \frac{t_0 - \lambda w_1/2}{Z_C}.$$

The symbol d in this formula is a local statistic; the hypergraph degree remains three throughout the argument.

Lemma 2.2. *Choose a uniform root and then condition on its exterior under the independent-set distribution. The resulting local model satisfies*

$$\mathbb{E}a = \alpha_H(\lambda), \qquad \mathbb{E}d = \mathbb{E}h = \mathbb{E}(r_0/Z_C) = \mathbb{E}(r_1/Z_C) = 0, \qquad \mathbb{E}\sigma = s.$$

Proof. The variable a is the conditional probability that the root is occupied. The conditional means of w_v, $X_v w_v$, and p_v are respectively $(w_0 + \lambda w_1)/Z_C$, $\lambda w_1/Z_C$, and t_0/Z_C. Averaging (1) proves the assertions for d and h. Equation (2) proves the assertions for r_y. Finally, exactly $2t$ roots have a star link. □

3 The counting bound

We seek an occupancy bound whose dependence on H is carried by the fixed number s. Put

$$A = \lambda + 4\lambda^2 + 2\lambda^3, \qquad B = 1 + 5\lambda + 9\lambda^2 + 3\lambda^3, \qquad u = \frac{A}{B}, \quad a_* = \frac{\lambda P'}{9P}. \tag{4}$$

The function u is an upper bound useful for twin-free hypergraphs, while a_* is the occupancy fraction of $H_{3,3}$.

Lemma 3.1. *For $0 \le \lambda \le 5/3$,*

$$\alpha_H(\lambda) \le \left(1 - \frac{3s}{2}\right) u(\lambda) + \frac{3s}{2} a_*(\lambda). \tag{5}$$

The proof uses the following explicit local inequality. Define

$$c_d = \frac{\lambda^2(\lambda+3)}{B}, \qquad c_h = -\frac{\lambda(1+\lambda)(\lambda+3)}{3B}, \qquad Q = 18\lambda^2 PB,$$

and the integer polynomials

$$\begin{aligned} N_0 &= 9\lambda^4 + 24\lambda^5 - 30\lambda^6 - 177\lambda^7 - 249\lambda^8 - 153\lambda^9 - 36\lambda^{10}, \\ N_1 &= 15\lambda^4 + 120\lambda^5 + 429\lambda^6 + 858\lambda^7 + 1044\lambda^8 + 807\lambda^9 \\ &\qquad + 384\lambda^{10} + 99\lambda^{11} + 9\lambda^{12}. \end{aligned} \tag{6}$$

Lemma 3.2. *Every local model in Section 2 satisfies, for $0 < \lambda \le 5/3$,*

$$u + \frac{3}{2}(a_* - u)\sigma - a + c_d d + c_h h + \frac{N_0 r_0 + N_1 r_1}{Q Z_C} \ge 0. \tag{7}$$

Proof. We give the finite enumeration and the exact positivity test. Label the vertices of each of the five link types. For every permitted marking, list all pairs and triples of link vertices meeting each marked pair in either zero or two vertices. Enumerate their subfamilies subject to the coordinate bounds $\deg_{\mathcal{F}}(i) \le 3 - \deg_L(i)$. For each subfamily, enumerate all U containing the required star center or marked endpoints. Finally, enumerate every $S \subseteq U$ and each legal root state to compute (3) and the two book statistics.

The implementation also records all twelve joint book–leaf states (b, x, y), where $b \in \{0, 1, 2\}$ is the number of occupied endpoints, $x \in \{0, 1\}$ is the leaf state, and $y \in \{0, 1\}$ is its availability. For each state, the incidence is counted with sign $+1$ at each endpoint and -2 at the leaf. Our r_0, r_1 are the entries $(2, 0, 0)$ and $(2, 0, 1)$. Recording the unused states makes the enumeration reproducible without incorporating the proposed inequality into the generation step.

Link	Marked edges	Constraint families	Counting vectors
$3K_2$	0	15453	369
$P_3 \sqcup K_2$	0	507	223
P_4	0	31	42
K_3	0	5	7
$3K_2$	1	828	140
$3K_2$	2	81	27
$3K_2$	3	8	4
$P_3 \sqcup K_2$	1	53	48
$K_{1,3}$	0	12	9

Table 1: Enumeration of the local models. Counting vectors may occur in more than one row.

Table 1 gives the sizes of the lists. In $3K_2$, markings with the same number of marked edges are equivalent under a link automorphism. After identical counting vectors are identified across the lists, there are 800 vectors. These include the star indicator, the number of marked edges, the five polynomials in (3), and all twelve transport polynomials.

For each vector, multiply the left side of (7) by $6QZ_C$. The result is an integer polynomial $S_C(\lambda)$ of degree at most 16. To specify the computation, put

$$U_0 = 18\lambda^2 PA, \quad V_0 = 3\lambda^2(\lambda P'B - 9PA), \quad C_0 = 18\lambda^4 P(\lambda+3), \quad D_0 = -6\lambda^3 P(1+\lambda)(\lambda+3).$$

Then the polynomial checked is

$$\begin{aligned} S_C = {} & 6(U_0 + V_0\sigma)Z_C - 6Q\lambda q_1 + C_0(6\lambda q_1 - w_0 - \lambda w_1) \\ & + D_0(6t_0 - 3\lambda w_1) + 6N_0 r_0 + 6N_1 r_1. \end{aligned} \tag{8}$$

The identities $U_0/Q = u$, $V_0/Q = 3(a_* - u)/2$, $C_0/Q = c_d$, and $D_0/Q = c_h$ verify the normalization.

For a nonzero polynomial $S_C(\lambda) = \sum_{j=0}^{D} a_j \lambda^j$, the coefficient of x^k in

$$3^D (1+x)^D S_C\left(\frac{5x}{3(1+x)}\right)$$

is the integer $\sum_{j=0}^{k} a_j 5^j 3^{D-j} \binom{D-j}{k-j}$. All these integers are nonnegative for all 800 vectors; 62 of the original polynomials vanish identically. This proves nonnegativity for $0 \le \lambda < 5/3$, and continuity gives the endpoint $\lambda = 5/3$. The positive factor $6QZ_C$ then proves (7).

The supplementary files `general/generate_views.py`, `general/verify.py`, and `general/interval_extension.py` perform these operations, starting from the five labeled links and the explicit coefficients (6). A second implementation enumerates all markings rather than using symmetry representatives and checks that the vectors and polynomials (8) agree. Neither verification uses floating-point arithmetic or an optimization solver. □

Proof of Lemma 3.1. For $0 < \lambda \le 5/3$, take expectations in (7). Lemma 2.2 cancels the four consistency terms and gives

$$\alpha_H \le u + \frac{3}{2}(a_* - u)s,$$

which is (5). At zero activity both occupancy fractions vanish. □

The next estimate supplies a strict loss from the first term of the interpolation. We retain a rational constant so that the last comparison can be checked without numerical integration.

Lemma 3.3. *The functions in* (4) *satisfy*

$$\int_0^1 \frac{A(\lambda)}{\lambda B(\lambda)}\,d\lambda < \frac{307}{500} < \frac{\log 253}{9} - \kappa.$$

Proof. Let $f(\lambda) = (1 + 4\lambda + 2\lambda^2)/B(\lambda)$, with its continuous value at zero. The numerator of f' is

$$-1 - 14\lambda - 35\lambda^2 - 24\lambda^3 - 6\lambda^4,$$

so f is decreasing on $[0, 1]$. Set $N = 1024$. The left Riemann sum gives

$$\int_0^1 f(\lambda)\,d\lambda \le \sum_{k=0}^{N-1} \frac{N^2 + 4kN + 2k^2}{N^3 + 5kN^2 + 9k^2N + 3k^3} < \frac{307}{500}. \tag{9}$$

The last inequality is an exact rational comparison included in the verifier. Thus it is a finite integer-arithmetic step, rather than an approximation to the integral.

The positive series for $\operatorname{arctanh}(1/3)$ gives

$$\log 2 > 2\left(\frac{1}{3} + \frac{1}{81} + \frac{1}{1215}\right) > \frac{693}{1000}.$$

Since $\log(256/253) < 3/253$, it follows that

$$\frac{\log 253}{9} > \frac{8(693/1000) - 3/253}{9} = \frac{307}{500} + \frac{259}{379500}.$$

Together with (9), this proves the claim. □

Proof of Theorem 1.1. Suppose $n > 0$. The number s is fixed by H, so we may divide (5) by λ and integrate from zero to one. Using $Z_H(0) = P(0) = 1$ and Lemma 3.3, we obtain

$$\frac{\log i(H)}{n} \le \left(1 - \frac{3s}{2}\right)\int_0^1 \frac{A}{\lambda B}\,d\lambda + \frac{3s}{2}\frac{\log 253}{9} \le \frac{\log 253}{9} - \kappa\left(1 - \frac{3s}{2}\right).$$

By Lemma 2.1, $n(1 - 3s/2) = n - 3t = q(H)$, giving the required bound. If $q(H) > 0$, the unpenalized comparison is strict.

It remains to determine equality when $q(H) = 0$. In that case, every edge belongs to a book. The non-twin vertices form a set of $t = n/3$ leaves, and the incidence graph between the t books and these leaves is a simple 3-regular bipartite graph. Choose each leaf independently with probability $1/2$. A book contributes four choices of endpoints if its three leaves are absent, and three choices otherwise. Consequently,

$$i(H) = 2^t \mathbb{E} \prod_{B \text{ a book}} f_B, \qquad f_B = 3 + \mathbf{1}_{\{\text{all three leaves of } B \text{ are absent}\}}.$$

Every leaf variable appears in three factors. Finner's inequality [3], the product-space form of Hölder's inequality, yields

$$\mathbb{E}\prod_B f_B \le \prod_B (\mathbb{E} f_B^3)^{1/3} = \left(\frac{4^3 + 7 \cdot 3^3}{8}\right)^{t/3} = \left(\frac{253}{8}\right)^{t/3}.$$

This recovers $i(H) \le 253^{t/3} = 253^{n/9}$ for full-book hypergraphs and allows us to examine equality directly.

Fix a leaf variable X and apply the three-factor Hölder inequality to it first. For the three incident books, replace f_B by $(\mathbb{E}_X f_B^3)^{1/3}$ and apply Finner's inequality to the remaining coordinates. The resulting upper bound is unchanged, since the third moment of each new

factor equals the third moment of the original factor. Hence equality in the final bound requires equality in this first conditional Hölder step for every assignment of the other leaves. Every such assignment has positive probability and every factor is positive.

For an incident book, the ratio $f_B(1)/f_B(0)$ is $3/4$ if its other two leaves are absent, and is one otherwise. Equality therefore forces the three incident books to have the same other two leaves. Indeed, the indicator that a pair of coordinates is simultaneously zero determines that pair: set one coordinate to one and all others to zero. Thus the three books have the same three-leaf neighborhood. Their degrees exhaust all incidences of those leaves, giving a $K_{3,3}$ component of the incidence graph and a copy of $H_{3,3}$ in H. Repeating the argument at every leaf proves the necessity. Conversely, each reference component has 253 independent sets, and counts multiply over components. □

4 Weighted bounds and stability

The local inequalities also give weighted bounds under additional structural assumptions. A hypergraph is linear if distinct edges intersect in at most one vertex. We first treat this class and hypergraphs without twins. We then prove a quantitative form of the partition-function bound for hypergraphs with an exact transversal.

For the linear case, put

$$A_{\mathrm{L}}(\lambda) = \lambda + 6\lambda^2 + 16\lambda^3 + 20\lambda^4 + 10\lambda^5 + 2\lambda^6,$$
$$B_{\mathrm{L}}(\lambda) = 1 + 7\lambda + 25\lambda^2 + 48\lambda^3 + 45\lambda^4 + 18\lambda^5 + 3\lambda^6.$$

The polynomials A and B retain their definitions from Section 3.

Theorem 4.1. *Let H be a nonempty simple 3-uniform 3-regular hypergraph on n vertices.*

1. *If H is linear, then, for every $\lambda > 0$,*

$$\alpha_H(\lambda) \le \frac{A_{\mathrm{L}}(\lambda)}{B_{\mathrm{L}}(\lambda)} < \frac{\lambda P'(\lambda)}{9P(\lambda)}, \qquad Z_H(\lambda) < P(\lambda)^{n/9}.$$

2. *If H has no pair of twins, then $\alpha_H(\lambda) \le A(\lambda)/B(\lambda)$ for every $\lambda \ge 0$. Moreover, for every $\lambda \ge 1$,*

$$Z_H(\lambda) \le P(\lambda)^{n/9} e^{-\kappa n}.$$

The rational function $A_{\mathrm{L}}/B_{\mathrm{L}}$ agrees with the $r = d = 3$ specialization of the bound of Cohen, Perkins, Sarantis, and Tetali [2, Proposition 11], proved there for cross-edge-free linear hypergraphs and $0 \le \lambda \le 1$. The first part of Theorem 4.1 allows all linear hypergraphs and all positive activities. Both parts follow from coefficientwise nonnegative local polynomials, so no restriction on the activity is needed for the occupancy bounds themselves.

Lemma 4.2. *Use the local polynomials of Section 2, and write $Z_C = q_0 + \lambda q_1$. Set*

$$C_{\mathrm{L}} = 6\lambda^2 + 24\lambda^3 + 30\lambda^4 + 12\lambda^5 + 2\lambda^6, \qquad D = 3\lambda + 7\lambda^2 + 2\lambda^3.$$

For every conditional local view of a linear 3-uniform 3-regular hypergraph, the polynomial

$$6A_{\mathrm{L}}Z_C - 6(B_{\mathrm{L}} - C_{\mathrm{L}})\lambda q_1 - C_{\mathrm{L}}(\lambda q_0' + \lambda^2 q_1') \tag{10}$$

has nonnegative coefficients. For every conditional local view whose link is not a star, the polynomial

$$12AZ_C - (12B - 6D)\lambda q_1 - D(w_0 + 2t_0) \tag{11}$$

has nonnegative coefficients.

Proof. We give the complete finite verification procedure. In the linear case the link is the matching $\{12, 34, 56\}$. A residual constraint can use at most one vertex of each matching edge. Thus there are twelve possible two-element constraints and eight possible three-element constraints. The family $\mathcal{F}$ of active constraints has maximum degree at most two, and distinct members of $\mathcal{F}$ intersect in at most one vertex. These restrictions follow respectively from regularity and linearity of the original hypergraph. Enumerate every subfamily of these twenty sets satisfying the restrictions, and independently choose every available set $U \subseteq [6]$. For each $S \subseteq U$, test whether it contains a member of $\mathcal{F}$, and, if not, whether it contains a matching edge. Counting these subsets by size gives q_0 and q_1.

This enumeration contains 2038 constraint families and 112 distinct pairs (q_0, q_1) after all choices of U. Expanding (10) for each pair gives nonnegative integer coefficients. The two identically zero polynomials arise from

$$(q_0, q_1) = ((1+\lambda)^3, (1+\lambda)^3) \quad \text{and} \quad (q_0, q_1) = ((1+\lambda)^6, (1+2\lambda)^3).$$

For the second assertion the link is one of $3K_2$, $P_3 \sqcup K_2$, P_4, and K_3. For each link L, enumerate all families $\mathcal{F} \subseteq \binom{V(L)}{2} \cup \binom{V(L)}{3}$ with $d_{\mathcal{F}}(u) \le 3 - d_L(u)$, followed by all available sets $U \subseteq V(L)$. The numbers of families are 15453, 507, 31, and 5, respectively. Direct subset counting gives 571 distinct triples $(q_0, q_1, w_0 + 2t_0)$. Expansion of (11) gives nonnegative integer coefficients in each case; three expansions are identically zero.

The treatment of singleton and repeated constraints is the same as in Section 2. Allowing U independently of the remaining degree budgets can add unrealizable views, but cannot omit an actual view. The verification therefore proves both assertions for every hypergraph in question. The supplementary directories `linear` and `twinfree` contain the complete coefficient lists and integer verification programs, together with independent enumerations. Only enumeration of finite set families, subset counting, and integer polynomial arithmetic are used. □

Proof of Theorem 4.1. Suppose first that H is linear. Conditional on a local view, set

$$a = \frac{\lambda q_1}{Z_C}, \qquad b = \frac{\lambda q_0' + \lambda^2 q_1'}{6Z_C}.$$

The quantity b is the average occupation probability of the six neighbors of the root. The graph joining vertices that share an edge of H is 6-regular, so a uniform root satisfies $\mathbb{E}a = \mathbb{E}b = \alpha_H(\lambda)$. Dividing (10) by $6B_{\mathrm{L}}Z_C$ and averaging therefore gives $\alpha_H \le A_{\mathrm{L}}/B_{\mathrm{L}}$. Direct expansion yields

$$\begin{aligned}\lambda P' B_{\mathrm{L}} - 9PA_{\mathrm{L}} = {} & 36\lambda^4 + 291\lambda^5 + 966\lambda^6 + 1713\lambda^7 + 1764\lambda^8 \\ & + 1053\lambda^9 + 324\lambda^{10} + 36\lambda^{11} > 0.\end{aligned}$$

Integrating the resulting strict occupancy comparison from 0 to λ proves the partition-function bound.

Now suppose that H has no twins. A star link would make its root and center twins, so only the four nonstar links occur. Define $b = (w_0 + 2t_0)/(6Z_C)$. The definitions of d and h give $b = a - d + h/3$; hence Lemma 2.2 gives $\mathbb{E}b = \mathbb{E}a = \alpha_H(\lambda)$. Dividing (11) by $12BZ_C$ gives

$$\frac{A}{B} \ge a - \frac{D}{2B}(a - b).$$

Taking expectations proves the asserted occupancy bound for $\lambda > 0$, and continuity gives the value at zero.

For $\lambda \ge 1$, the reference occupancy is strictly larger than A/B. Indeed,

$$\lambda P' B - 9PA = -9\lambda^4 - 6\lambda^5 + 51\lambda^6 + 81\lambda^7 + 36\lambda^8$$

has, after substituting $\lambda = 1 + x$, the positive coefficient list

$$(153, 1095, 3360, 5775, 6081, 4017, 1626, 369, 36).$$

Lemma 3.3 gives a deficit of at least κ at $\lambda = 1$. Integrating the strict occupancy comparison from 1 to λ preserves this deficit and proves the second assertion. □

An *exact transversal* is a vertex set meeting every edge in exactly one vertex. Under this assumption we can detect not only local defects, but every component that differs from the reference hypergraph. Let $b(H)$ denote the number of vertices of H that do not belong to a connected component isomorphic to $H_{3,3}$.

Theorem 4.3. *Let H be a nonempty simple 3-uniform 3-regular hypergraph on n vertices with an exact transversal. For every $\lambda \geq 1$,*

$$Z_H(\lambda) \leq P(\lambda)^{n/9} \exp\left(-\frac{b(H)}{36864(1+\lambda)^{203}}\right).$$

In particular, equality in $Z_H(\lambda) \leq P(\lambda)^{n/9}$ holds if and only if H is a disjoint union of copies of $H_{3,3}$.

For fixed $\lambda \geq 1$, a sequence with $\log Z_H(\lambda) \geq (n/9) \log P(\lambda) - o(n)$ consequently has $b(H) = o(n)$. Thus all but $o(n)$ vertices already lie in complete reference components. The explicit constant in the theorem is chosen for the proof and is not optimized.

We use the following form of Finner's inequality [3]. If nonnegative functions on a product probability space each depend on a specified set of coordinates, and every coordinate belongs to at most k of these sets, then

$$\mathbb{E} \prod_a f_a \leq \prod_a (\mathbb{E} f_a^k)^{1/k}.$$

An exact transversal gives a factorization in which the coordinates belong to either two or three factors. Applying this inequality to the two classes successively gives a mixed norm. The following lemma records the precise bound, including a loss for all but two local factor types.

Lemma 4.4. *Let T be an exact transversal of H, put $m = |T| = n/3$, and fix $\lambda \geq 1$. There are independent Bernoulli coordinates of parameter $p = \lambda/(1+\lambda)$ and positive factors G_a, indexed by $a \in T$, such that each coordinate occurs in at most three factors and*

$$\begin{aligned} Z_H(\lambda) &\leq (1+\lambda)^{2m-z} \mathbb{E} \prod_{a \in T} G_a, \\ (1+\lambda)^{2m-z} \prod_{a \in T} (\mathbb{E} G_a^3)^{1/3} &\leq P(\lambda)^{n/9} (199/200)^{h/3}. \end{aligned} \tag{12}$$

Here z counts the star links rooted in T, and h counts the factors other than the following two types, called good factors. A matching factor has support consisting of three disjoint pairs and is

$$f = 1 + \lambda \, \mathbf{1}_{\text{no pair is fully selected}}.$$

A star factor has a three-element support and is

$$f = 1 + 2\lambda + \lambda^2 \mathbf{1}_{\text{all three coordinates are zero}}.$$

For a good factor, $G_a = f$, and every coordinate in its support occurs in exactly three factors. Every factor has at most six coordinates and satisfies $(1+\lambda)^{-1} \leq G_a \leq (1+\lambda)^3$.

The construction and the finite mixed-norm verification are given in Appendix A. The remaining argument is an analytic stability estimate; it does not require additional computation.

Proof of Theorem 4.3. Apply Lemma 4.4, and put $L = 1 + \lambda$. For a coordinate x incident only with good factors, consider the three ratios $f(x = 1)/f(x = 0)$ as functions of the other coordinates. Call x incompatible if these functions are not all identical, and let v be the number of incompatible coordinates. We first claim that

$$b(H) \le 39h + 9v. \tag{13}$$

For a matching factor, its ratio differs from 1 precisely when the partner of x is selected and neither of the other two pairs is fully selected; the nonunit value is $1/L$. For a star factor, the event is that its other two coordinates vanish, and the nonunit value is $(1 + 2\lambda)/L^2$. These two values differ. Equality of two matching ratio functions identifies first the partner of x, and then the two remaining pairs as the minimal forbidden sets. Equality of two star ratio functions identifies their other two coordinates. Consequently, if a coordinate is incident only with good factors and is compatible, its three incident factors have the same type and support. Since every coordinate has multiplicity three, these factors form a closed copy of $H_{3,3}$, after restoring the star centers when necessary.

A bad factor has at most six coordinates, each shared with at most two other factors. At most $12h$ good factors are therefore adjacent to a bad factor. Every other good factor outside a reference component must be incident with an incompatible coordinate, accounting for at most $3v$ further factors. Each connected component of H has three times as many vertices as transversal vertices, by counting its edges. Thus $b(H) \le 3(h + 12h + 3v)$, proving (13).

For an incompatible coordinate x, the union T_x of the supports of its three good factors has size at most 16. Fix an assignment to $T_x \setminus \{x\}$ on which two ratios differ. We show that on this assignment the one-coordinate Hölder ratio satisfies

$$R_x := \frac{\mathbb{E}_x f_1 f_2 f_3}{\prod_{i=1}^{3} (\mathbb{E}_x f_i^3)^{1/3}} \le 1 - \delta, \qquad \delta = \frac{1}{16L^8}. \tag{14}$$

Each factor ratio is one of 1, $1/L$, and $(1 + 2\lambda)/L^2$; distinct values differ by at least $1/(2L)$. Define probability distributions $Q_i(t) = \Pr(x = t) f_i(t)^3 / \mathbb{E}_x f_i^3$, for $t \in \{0, 1\}$. For factor ratio r, its mass at 1 is $\psi(r) = pr^3/(1 - p + pr^3)$. On $[1/L, 1]$,

$$\psi'(r) = \frac{3p(1-p)r^2}{(1 - p + pr^3)^2} \ge \frac{3}{2L^3}.$$

Thus two of the distributions have total variation distance at least $3/(4L^4)$. Put $B_0 = \sum_{t=0}^{1} \sqrt{Q_1(t) Q_2(t)}$. The Cauchy–Schwarz inequality gives $1 - B_0 \ge \|Q_1 - Q_2\|_{\mathrm{TV}}^2/2$. Hölder's inequality, first grouping the first two factors, gives $R_x \le B_0^{2/3}$. It follows that

$$1 - R_x \ge \frac{2}{3}(1 - B_0) \ge \frac{1}{3}\|Q_1 - Q_2\|_{\mathrm{TV}}^2 \ge \frac{3}{16L^8} \ge \delta,$$

as required.

Every coordinate belongs to at most 16 of the sets T_x: a center whose set contains it must share one of its at most three factors. Each T_x consequently intersects at most 256 such sets. Greedily choose $k \ge v/256$ pairwise disjoint sets T_x. No factor contains two chosen centers. Integrating these centers therefore replaces their factors by $G'_a = (\mathbb{E}_x f_a^3)^{1/3}$ and gives the exact identity

$$\mathbb{E} \prod_a G_a = \mathbb{E}\left(\prod_a G'_a \prod_x R_x \right),$$

where unaltered factors retain the notation $G'_a = G_a$ and the remaining expectation is over all unintegrated coordinates. Each G'_a still lies between L^{-1} and L^3.

Weight the remaining product measure by $\prod_a G'_a$ and normalize it to obtain a probability measure μ. The chosen witness for x fixes at most 15 coordinates and affects at most 45

factors. Conditional on every coordinate outside that block, its probability under μ is at least $L^{-15-4\cdot45} = L^{-195}$. Here each prescribed Bernoulli value has probability at least $1/L$, and the ratio between the maximum and minimum of each affected factor is at most L^4. The witness blocks are disjoint. Successive conditioning on their complements, using (14), therefore gives

$$\mathbb{E}_\mu \prod_x R_x \le (1-\eta)^k, \qquad \eta = \delta L^{-195} = \frac{1}{16L^{203}}.$$

This step uses a uniform conditional probability bound, not independence of the witnesses under μ.

Finner's inequality on the remaining coordinates yields $\mathbb{E}\prod_a G'_a \le \prod_a (\mathbb{E}G_a^3)^{1/3}$; integrating one coordinate of a good factor has preserved its third moment. Substituting this and the last estimate into (12) gives

$$Z_H(\lambda) \le P(\lambda)^{n/9}(199/200)^{h/3}(1-\eta)^k \le P(\lambda)^{n/9}\exp\left(-\frac{h}{600} - \frac{\eta v}{256}\right).$$

Set $\kappa(\lambda) = \eta/2304 = 1/(36864L^{203})$. Since $39\kappa(\lambda) \le 1/600$ and $9\kappa(\lambda) = \eta/256$, (13) proves the theorem. The equality assertion follows because $b(H) = 0$ precisely for disjoint unions of reference components, whose partition functions multiply. □

We now remove the transversal assumption on the interval $[2/3, 5/3]$. The first step is a stability estimate for full-book hypergraphs, meaning that every edge contains a twin pair. Its proof uses only the product formula and a strict form of the two-point Hölder inequality.

Lemma 4.5. *Put $\gamma = 2^{-54}$. Every full-book 3-uniform 3-regular hypergraph satisfies*

$$Z_H(\lambda) \le P(\lambda)^{n/9} e^{-\gamma b(H)} \qquad (2/3 \le \lambda \le 5/3).$$

Proof. There are $m = n/3$ books and m leaves. Put $L = 1+\lambda$ and choose each leaf independently with probability $p = \lambda/L$. Summing over the two endpoints of each book gives

$$Z_H(\lambda) = L^m \mathbb{E}\prod_B f_B, \qquad f_B = 1 + 2\lambda + \lambda^2 \mathbf{1}_{\{\text{all three leaves of } B \text{ are absent}\}}.$$

Each leaf belongs to three factors, and $L^m \prod_B (\mathbb{E}f_B^3)^{1/3} = P(\lambda)^{n/9}$. Indeed, $L^3\mathbb{E}f_B^3 = L^6 + (L^3 - 1)(1+2\lambda)^3 = P(\lambda)$.

At a leaf x, compare the three functions $f_B(x=1)/f_B(x=0)$. Their values are one and $r = (1+2\lambda)/L^2$; the latter occurs exactly when the other two leaves of the book are absent. If the functions agree identically, their other two leaves coincide. The three books then form a complete reference component. Thus every leaf outside reference components admits an assignment witnessing two different ratios. There are $v = b(H)/3$ such leaves. The union T_x of the supports of its three factors has at most seven coordinates. Each coordinate belongs to at most seven such sets, so each T_x intersects at most 49 of them, including itself. Choose $k \ge v/49 = b(H)/147$ pairwise disjoint sets T_x.

We record a uniform loss at a chosen center x under its witness assignment. On the stated activity interval,

$$p, 1-p \ge 3/8, \qquad 1/2 \le r \le 7/8, \qquad \frac{\max f_B}{\min f_B} \le 2.$$

For the three incident factors, define

$$R_x = \frac{\mathbb{E}_x f_1 f_2 f_3}{\prod_{i=1}^3 (\mathbb{E}_x f_i^3)^{1/3}}, \qquad Q_i(t) = \frac{\Pr(x=t) f_i(t)^3}{\mathbb{E}_x f_i^3} \quad (t \in \{0,1\}).$$

A factor with ratio u has $Q_i(1) = h(u)$, where $h(u) = pu^3/(1-p+pu^3)$. For $1/2 \le u \le 1$, one has $h'(u) \ge 27/256 > 1/16$. Two distinct ratios differ by at least $1/8$, so their distributions have

total variation distance at least $1/128$. Writing $\beta = \sum_t \sqrt{Q_1(t)Q_2(t)}$, Cauchy's inequality gives $1 - \beta \geq \|Q_1 - Q_2\|_{\mathrm{TV}}^2/2$, and Hölder gives $R_x \leq \beta^{2/3}$. Consequently, on the witness event,

$$1 - R_x \geq \frac{1}{3}\|Q_1 - Q_2\|_{\mathrm{TV}}^2 \geq \frac{1}{49152} > 2^{-16}.$$

Always $0 < R_x \leq 1$.

Integrate the k chosen centers first, replacing each incident factor by $g_B = (\mathbb{E}_x f_B^3)^{1/3}$. No factor contains two chosen centers. Therefore

$$\mathbb{E}\prod_B f_B = \mathbb{E}\left(\prod_B g_B \prod_x R_x\right),$$

where unchanged factors are also denoted by g_B. Reweight the remaining product measure by $\prod_B g_B$ and normalize it to obtain μ. Each witness fixes at most six coordinates and affects at most 18 factors. The pointwise ratio of every factor's maximum to minimum remains at most two. Conditional on all coordinates outside the witness block, its probability under μ is therefore at least $4^{-6}2^{-18} = 2^{-30}$. The witness blocks are disjoint. Successive conditioning gives $\mathbb{E}_\mu \prod_x R_x \leq (1 - 2^{-46})^k$. Finally, Finner's inequality applies to the remaining coordinates, and the replacement preserves every third moment. Hence

$$Z_H(\lambda) \leq P(\lambda)^{n/9}(1 - 2^{-46})^k \leq P(\lambda)^{n/9} \exp\left(-\frac{b(H)}{147 \cdot 2^{46}}\right).$$

Since $147 < 2^8$, this implies the claimed value of γ. □

Proof of Theorem 1.2. We first extend the local-defect estimate throughout the interval. Define

$$\Delta(\lambda) = \frac{\log P(\lambda)}{9} - \int_0^\lambda \frac{A(t)}{tB(t)}\,dt.$$

The interpolation bound, with s fixed, gives

$$\log Z_H(\lambda) \leq \frac{n}{9}\log P(\lambda) - q(H)\Delta(\lambda) \qquad (0 \leq \lambda \leq 5/3).$$

We claim that $\Delta(\lambda) > 1/10000$ for $2/3 \leq \lambda \leq 5/3$. Multiplying Δ' by $9\lambda PB$ gives

$$\lambda P'B - 9PA = \lambda^4(-9 - 6\lambda + 51\lambda^2 + 81\lambda^3 + 36\lambda^4).$$

The coefficients of the polynomial in parentheses after the shift $\lambda = 2/3 + x$ are

$$(367/9, 638/3, 309, 177, 36).$$

Thus Δ is increasing for $\lambda \geq 2/3$.

To check the left endpoint exactly, use the decreasing function f from Lemma 3.3. Put $N = 2048$ and

$$U = \frac{2}{3N}\sum_{j=0}^{N-1} f\left(\frac{2j}{3N}\right), \qquad T(z) = 2\sum_{j=0}^{19}\frac{z^{2j+1}}{2j+1}.$$

Then $\int_0^{2/3} f \leq U$, while $P(2/3) = 5471/81$ and the positive series for $\log((1+z)/(1-z))$ yield

$$\Delta(2/3) > \frac{6T(1/3) + T(287/10655)}{9} - U > \frac{1}{10000}.$$

The final comparison is a finite rational calculation, reproduced by two independent supplementary programs. We have proved that the logarithmic deficit $D = (n/9)\log P(\lambda) - \log Z_H(\lambda)$ satisfies $D \geq q(H)/10000$.

For each edge $e = \{a, b, c\}$ containing no twin pair, add two new vertices u_e, v_e and replace e by $\{u_e, v_e, a\}, \{u_e, v_e, b\}, \{u_e, v_e, c\}$. Call the resulting hypergraph $\widetilde{H}$, and write $q = q(H)$. The new vertices have degree three, and every old vertex retains its degree. The old books are unchanged, so $\widetilde{H}$ is simple, 3-uniform, 3-regular and full-book, with $n + 2q$ vertices. Each independent set of H extends to $\widetilde{H}$ by choosing at most one endpoint of each new pair. Summing their weights gives

$$Z_{\widetilde{H}}(\lambda) \geq (1 + 2\lambda)^q Z_H(\lambda).$$

A reference component of $\widetilde{H}$ containing no new vertex was already a reference component of H. Every other reference component contains a whole new pair, and hence at most seven old vertices; there are at most q such components. Thus $b(\widetilde{H}) \geq b(H) - 7q$.

Apply Lemma 4.5 to $\widetilde{H}$. With $\gamma = 2^{-54}$, this gives

$$D \geq \gamma b(H) - q\left(\frac{2}{9}\log P(\lambda) - \log(1 + 2\lambda) + 7\gamma\right) \geq \gamma b(H) - q.$$

For the last step, $P(\lambda) \leq (1 + \lambda)^9$ and $(1 + \lambda)^2/(1 + 2\lambda) \leq 2$ on our interval, so the coefficient of q is at most $\log 2 + 7\gamma < 1$. Combining this with $q \leq 10000D$ gives $D \geq \gamma b(H)/10001$. We may therefore take $c = 1/(10001 \cdot 2^{54})$. If $b(H) > 0$ the unpenalized bound is strict; if $b(H) = 0$, the partition functions of the reference components multiply. □

Declaration of AI use

The author used GPT-6.1 to assist with exploring proof strategies, checking calculations, and improving the exposition. The author takes responsibility for the content of this paper.

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A The mixed-norm certificate

We prove Lemma 4.4. The proof first expresses the partition function using factors on a product probability space. There are only 93 possible marked factor types. Two satisfy the required bound identically; a rational interval certificate proves a uniform strict bound for the other types on the whole interval $1 \leq \lambda < \infty$.

Fix an exact transversal T, put $m = |T|$, and let $B_0 = V(H) \setminus T$. Counting edges gives $|B_0| = 2m$. For $a \in T$, let L_a be its link on B_0. Each L_a is a simple three-edge graph without isolated vertices. A vertex of B_0 belongs to only one link if and only if it is the center of a star

link. Such a center occurs in no other link, and different star links have different centers. Let z be the number of star links and remove their centers to form B', of size $2m - z$.

For a nonstar transversal vertex a, let J_a indicate that the selected coordinates contain no edge of L_a. Summing over a gives $f_a = 1+\lambda J_a$. For a star link, sum simultaneously over its transversal vertex and its center; the resulting factor is $f_a = 1 + 2\lambda + \lambda^2 \mathbf{1}_{\text{all three leaves are zero}}$. Let each coordinate of B' independently equal 1 with probability $p = \lambda/(1 + \lambda)$. The partition-function sum is then exactly

$$Z_H(\lambda) = (1 + \lambda)^{2m-z} \mathbb{E} \prod_{a \in T} f_a. \tag{15}$$

The product measure here rewrites the sum; it is not an independence assertion about the hard-core measure on H.

Every vertex of B' has either two or three distinct transversal neighbors. Partition B' accordingly into B_2 and B_3. A vertex of B_2 has local degrees 2 and 1 in its two links, whereas a vertex of B_3 has local degree 1 in each of three links. For each $a \in T$, let U_a and W_a be the coordinates of its factor in B_2 and B_3, respectively. Let D_a be the vertices of degree two in L_a when the link is nonstar, and put $D_a = \varnothing$ for a star. In particular, $D_a \subseteq U_a$. Define

$$\omega = \left(\frac{\lambda + 3}{2}\right)^{1/6}, \qquad \rho = \frac{41}{50}\omega^3, \qquad q_a = |D_a| - |U_a \setminus D_a|,$$

and

$$F_a = f_a \omega^{\sum_{b \in D_a} X_b - \sum_{b \in U_a \setminus D_a} X_b}.$$

Each B_2 coordinate has exponent $+X_b$ in one factor and $-X_b$ in the other. Hence $\prod_a F_a = \prod_a f_a$ pointwise and $\sum_a q_a = 0$.

First fix the B_3 coordinates and apply Finner's inequality with exponent two to B_2. Then set

$$G_a = (\mathbb{E}_{U_a} F_a^2)^{1/2}, \qquad K_a = \mathbb{E}_{W_a} G_a^3 = \mathbb{E}_{W_a} (\mathbb{E}_{U_a} F_a^2)^{3/2}.$$

Equation (15) gives the first inequality of (12). Each B_3 coordinate occurs in three factors, so a further application of Finner gives $\mathbb{E} \prod_a G_a \le \prod_a K_a^{1/3}$. Also $1 \le \omega^6 \le 1 + \lambda$, the exponent of ω in F_a has absolute value at most six, and $1 \le f_a \le (1 + \lambda)^2$. Thus $(1 + \lambda)^{-1} \le F_a, G_a \le (1 + \lambda)^3$.

Put $C(\lambda) = P(\lambda)/(1 + \lambda)^6$. The local inequalities to be certified are

$$K_a \le C(\lambda)(1 + \lambda)^{3\varepsilon_a} \rho^{q_a}, \tag{16}$$

where $\varepsilon_a = 1$ for a star and 0 otherwise. Except for a matching with all six coordinates in B_3 or a star with all three leaves in B_3, the right-hand side can be multiplied by 199/200. In the two exceptions $U_a = \varnothing$, $G_a = f_a$, and direct evaluation gives, respectively,

$$K_a = 1 + \left((1 + \lambda)^3 - 1\right) \left(\frac{1 + 2\lambda}{(1 + \lambda)^2}\right)^3 = C(\lambda),$$

$$K_a = (1 + 2\lambda)^3 + \frac{(1 + \lambda)^6 - (1 + 2\lambda)^3}{(1 + \lambda)^3} = (1 + \lambda)^3 C(\lambda).$$

The other local types are obtained by taking one of $P_4, K_3, P_3 \sqcup K_2, 3K_2, K_{1,3}$ and marking each of its degree-one vertices as belonging to B_2 or B_3. Degree-two vertices must belong to B_2, and the star center has already been integrated out. The numbers of marked types are $4, 1, 16, 64, 8$, giving 93 types in total.

We describe the integer certificate for the 91 nonexceptional types. Its explicit formulas also specify the finite verification independently of the accompanying program. Substitute

$$t = \omega^{-1}, \qquad a = 2 - 3t^6, \qquad b = 2 - 2t^6, \qquad d = 4 - 5t^6.$$

Then $\lambda = a/t^6$, $p = a/b$, $1-p = t^6/b$, and $\rho = 41/(50t^3)$. The full activity range is $0 < t \le 2^{-1/6} < 891/1000$. All interval checks below take place on the larger closed interval $0 \le t \le 891/1000$, after cancellation of negative powers of t.

Fix a marked type, suppress its subscript, and write $u = |U|$, $w = |W|$, and $\varepsilon \in \{0,1\}$ for its star indicator. For an assignment $y \in \{0,1\}^W$ with j selected coordinates, enumerate all $x \in \{0,1\}^U$. Let k be the number selected in x, let $g = \sum_{b\in D} x_b - \sum_{b\in U\setminus D} x_b$, and let $I \in \{0,1\}$ be the indicator in f. Define

$$v = 6(u-k) - 2g - 12(1+\varepsilon) + 12(1-I), \qquad \beta = 2(1+\varepsilon)I, \qquad \eta = 2\varepsilon(1-I).$$

The nonstar factor equals $(t^6 + aI)/t^6$. The star factor equals b^2/t^{12} when $I = 1$, and $t^6 d/t^{12}$ when $I = 0$. Consequently,

$$\mathbb{E}_U F^2 = b^{-u} \sum_x t^v a^k b^\beta d^\eta.$$

For each fixed y, let v_y be the smallest occurring exponent v, and define

$$B_y(t) = \sum_x t^{v-v_y} a^k b^\beta d^\eta, \qquad e_y = 6(w-j) + \frac{3}{2}v_y + 3q + 18\varepsilon.$$

All v_y are even, all e_y are nonnegative, and all powers in B_y are nonnegative integers; these assertions are checked while enumerating the 93 types. Finally put

$$T(t) = \sum_{k=0}^{6} p_k t^{36-6k} a^k, \qquad (p_0,\dots,p_6) = (1,9,36,75,81,42,9).$$

Substitution into the mixed norm gives the exact normalized expression

$$\frac{K}{C(\lambda)(1+\lambda)^{3\varepsilon}\rho^q} = \frac{b^{6-w-3u/2-3\varepsilon}}{T(t)(41/50)^q} \sum_y t^{e_y} a^j B_y(t)^{3/2}. \tag{17}$$

The functions a, b, d are strictly positive throughout the checked interval, as is T. Thus the right-hand side extends continuously to $t = 0$. The 91 expressions have 25 distinct forms after identical summands and identical full expressions are merged.

For completeness, the interval algorithm uses scaled integer endpoints with scale $M = 2^{60}$. An interval is represented by integers (l, u) and denotes $[l/M, u/M]$. Addition and subtraction use their usual endpoint formulas. For nonnegative intervals, multiplication replaces the endpoints by

$$\left(\left\lfloor \frac{l_1 l_2}{M} \right\rfloor, \left\lceil \frac{u_1 u_2}{M} \right\rceil \right),$$

and division, when $l_2 > 0$, uses $(\lfloor l_1 M/u_2 \rfloor, \lceil u_1 M/l_2 \rceil)$. Square roots use $(\lfloor\sqrt{lM}\rfloor, \lceil\sqrt{uM}\rceil)$, computed by integer square roots. Negative powers of a positive interval are evaluated by reciprocation. These operations give outward enclosures at every step of (17).

Start with the enclosing dyadic interval for $[0, 891/1000]$. For each expression, accept an interval if its upper bound is at most $\lfloor (199/200)M \rfloor / M$; otherwise bisect it. The certificate has 963 nodes and 3510 accepted expression–interval pairs, with maximum depth 15. For each of the 25 expressions, the accepted intervals cover the initial interval and satisfy the stated bound. The directory `supplement/transversal` contains the generator, all accepted intervals and bounds, and an independent verifier that rebuilds the expressions at scale 2^{90}. This proves (16) with factor $199/200$ for every nonexceptional type throughout the activity interval.

Multiplying these local inequalities over the m transversal vertices, using $\sum_a q_a = 0$ and $\sum_a \varepsilon_a = z$, now gives

$$\prod_a K_a \le (1+\lambda)^{3z} C(\lambda)^m (199/200)^h.$$

After taking cube roots and multiplying by $(1+\lambda)^{2m-z}$, the powers of $1+\lambda$ cancel because $C = P/(1+\lambda)^6$. The resulting bound is $P^{m/3}(199/200)^{h/3}$, which is the second inequality in (12). This completes the proof of Lemma 4.4.